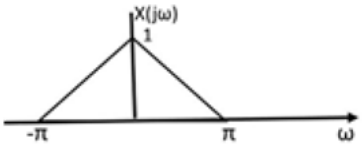
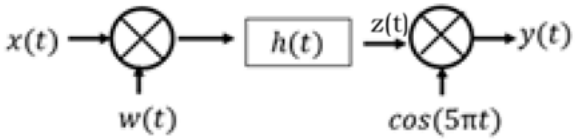
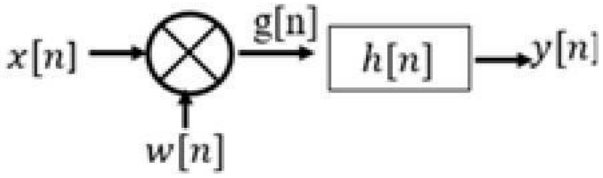
	SIGNALS AND SYSTEMS JULY-DEC 2025 ASSIGNMENT 3 Last Date for Submission: Oct 27 Answer any 5 (write and upload in LMS)
1	Consider the system depicted in the figure below and let the Fourier transform of the input signal $x(t)$ be $X(j\omega)$. Sketch $Z(j\omega)$ and $Y(j\omega)$ for $w(t) = \cos(5\pi t)$ and $h(t) = \frac{\sin(6\pi t)}{\pi t}$.  
2	Consider the discrete time system depicted in the figure below. Let $h[n] = \frac{\sin(\frac{n\pi}{2})}{n\pi}$. Determine the output response $y[n]$ for the input, $x[n] = \delta[n] - \frac{\sin(\frac{n\pi}{2})}{n\pi}$, and $w[n] = (-1)^n$. Also sketch $G(e^{j\Omega})$ and $Y(e^{j\Omega})$. 
3	A stable system has the indicated input $x(t)$ and output $y(t)$. Determine the transfer function and impulse response of the system. (i). $x(t) = e^{-t}u(t), y(t) = e^{-2t}\cos(t)\,u(t)$ (ii). $x(t) = e^{-2t}u(t), y(t) = -2e^{-t}u(t) + 2e^{-3t}u(t)$
4	Certain causal discrete time LTI system has frequency response $H(e^{j\Omega}) = \frac{1-\frac{1}{2}e^{-j\Omega}+\frac{1}{4}e^{-j2\Omega}+\frac{1}{8}e^{-3j\Omega}}{(1-\frac{1}{4}e^{-j\Omega})(1-\frac{1}{4}e^{-j2\Omega})}$. Determine impulse response and write the difference equation for this system.

5	Given two signals, $x[n] = \{1,2,3,\underset{\uparrow}{0},3,2,1\}$ and $y[n] = \{-3,-2,-1\ \underset{\uparrow}{0},\ 1,\ 2,\ 3\}$. Convolve two sequences using the Fourier analysis.
6	Use table of transforms and properties to find the inverse FT of the following signals a. $X(jw) = \frac{jw}{(2+jw)^2}$ b. $X(jw) = \frac{4\sin(2w-4)}{2w-4} - \frac{4\sin(2w+4)}{2w+4}$
7	Use table of transforms and properties to find DTFT of the following signals a. $x[n] = \left(\frac{1}{3}\right)^n u[n+2]$ b. $x[n] = \cos(\frac{\pi n}{4})\left(\frac{1}{2}\right)^n u[n-2]$
8	Use the tables of transform and properties to find FT of the following signals a. $x(t) = \sin(2\pi t)e^{-t}u(t)$ b. $x(t) = te^{-3\lvert t-1\rvert}$
9	Find FS coefficient by method of inspection for $x(t) = 3\cos(\frac{\pi t}{4} + \pi/6)$
10	Determine the DTFS coefficient of the following signal using method of inspection a. $x[n] = \cos\left(\frac{\pi}{3}n\right) + \sin(\frac{\pi}{4}n)$